AMAN MISHRA

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TECHNICAL SKILLS

Languages
Python

Tools & Concepts
REST APIs, Docker, Git, Web Scraping (Crawl4AI, Trafilatura), Tavily Search API, Celery, Redis

Frameworks & Libraries
Django Ninja, LangGraph, LangChain, Scikit-learn, SQLAlchemy

AI Observability & Tracing
LangSmith

AI / ML
Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic Workflows, Hybrid Retrieval, Reranking, Embeddings

Databases
PostgreSQL

PROJECTS

AI Receptionist Agent , *Live Link: https://ai-receptionist-e48c.onrender.com/api_v1/docs*
- Built and deployed an AI Receptionist for dental clinics using LangGraph, automating appointment booking, cancellation, rescheduling, FAQ handling, and emergency escalation.
- Engineered a stateful scheduling workflow with PostgreSQL-backed memory, validation pipelines, and automated IST→UTC conversion for Cal.com integrations.
- Implemented Hybrid RAG with pgvector semantic search and caching, reducing repeated FAQ latency from ~2.8s to ~1.0s while maintaining grounded responses.
- Designed fault-tolerant workflows with automated supervisor email escalation during calendar API failures.
- Integrated Twilio WhatsApp Webhooks, Cal.com scheduling APIs, LangSmith tracing, and PostgreSQL persistence to support production-grade conversational appointment management.

Newsletter Agent , *Subscribe here: https://newsletter-yend.onrender.com/*
- Built a LangGraph workflow that researches geopolitical events, ranks source relevance, generates newsletters, validates outputs, and delivers reports via email.
- Used LangSmith traces to identify browser scraping as the primary bottleneck; replaced it with Trafilatura HTML parsing, reducing latency ~80% (95s → 15s) and improving P99 tail latency 6x. Cut token usage ~70% (40k → 13k) via content filtering and context-size optimisation, lowering inference cost
- Implemented an LLM reflection loop to validate generated newsletters against source articles before delivery.
- Scheduled cron job for automated Newsletter delivery via Brevo Email API.

Meeting Analyser , *Live Link: https://arvya-meeting-analyser.onrender.com/api_v1/docs*
- Designed and deployed a LangGraph-based workflow engine for meeting intelligence, modeling report generation, persistence, and delivery as extensible processing nodes.
- Built a reusable transcript-processing pipeline supporting multiple ingestion paths while maintaining a single analysis layer, reducing workflow duplication and simplifying future integrations.
- Implemented schema-driven LLM extraction with Qwen3-32B and Groq Whisper, converting unstructured audio and transcript data into structured business intelligence artifacts.
- Developed resilient file-processing infrastructure using UUID-prefixed storage, automatic resource cleanup, database persistence, and workflow-level error handling.
- Deployed a production backend on Render with PostgreSQL (Supabase), recipient management APIs, and automated email distribution through Brevo.

RAG Powered Job Matching Agent
- Built an agentic job search pipeline using LangGraph & Django Ninja that autonomously finds, scrapes, and ranks job listings
- Designed a Hybrid RAG pipeline (semantic search + keyword extraction + Voyage reranking) to score resume-job fit from 0–100
- Implemented intent routing to classify queries into search or chat flows with automatic query rewriting from chat history
- Built a self-correcting reflection loop (max_iter=1) to detect LLM hallucinations before surfacing results
- Engineered stateful persistence via PostgreSQL Checkpointer, preserving session memory across server restarts
- Integrated Crawl4AI for deep job scraping with a 3-retry + query-rewrite fallback on failed searches
- Offloaded scraping and LLM inference to Celery + Redis workers for a non-blocking API layer

EDUCATION

B.E. Computer Science & Engineering, 2027 | Bhopal
Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV)